
Mean Flows as Controllable Priors for Solving Inverse Problems

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Abstract

1 Training-free inverse solvers built on flow priors repeatedly need to predict the final
2 clean sample reached from an intermediate generative state. Standard Flow Match-
3 ing models expose only instantaneous velocity, so distant clean-state predictions
4 require either costly numerical integration or coarse extrapolation. We instead use
5 MeanFlow models as controllable priors, leveraging their learned finite-interval
6 transport maps to predict clean states directly across arbitrary time horizons. This
7 interface supports existing training-free solvers without retraining and motivates
8 *Receding-Horizon State Optimization* (RHSO), which optimizes the current state
9 through its predicted clean sample, advances one generative interval, and replans.
10 An idealized analysis shows that intervention time changes the prior-induced con-
11 trol geometry along the trajectory. Across five ImageNet restoration tasks, the
12 pMF MeanFlow system is competitive with the JiT standard-flow system under
13 matched solver budgets and stronger at small trajectory budgets. RHSO is the
14 strongest solver we evaluate on MeanFlow, achieving the best five-task mean SSIM
15 and LPIPS. Matched standard-flow controls indicate that most of this gain comes
16 from distributing optimization across the trajectory rather than deepening endpoint
17 prediction, while ablations reveal a trade-off between reconstruction quality and
18 measurement consistency under excessive control.

19 1 Introduction

20 Inverse problems seek to recover an unknown signal x^* from an indirect, incomplete, or noisy
21 observation

$$y = A(x^*) + \eta, \quad (1)$$

22 where A is a known forward operator and $\eta \sim \mathcal{N}(0, \sigma^2 I)$ is measurement noise. Because A can
23 discard information, measurement consistency alone need not identify a unique or perceptually
24 plausible solution. Maximum-a-posteriori recovery maximizes $p(x | y) \propto p_{Y|X=x}(y | x) p_X(x)$, but
25 the prior over natural signals is intractable, so one can minimize a proxy summing a data-fidelity term
26 with a regularizer standing in for $-\log p_X(x)$; deep generative models are effective instantiations of
27 that regularizer [1–5].

28 Continuous normalizing flows [6] trained with Flow Matching (FM) objectives, particularly when
29 paired via (minibatch) optimal transport [7–9], are now a ubiquitous generative modeling tool and
30 scale to high-resolution synthesis [10–13]. They learn a velocity field $v_\theta(x, t)$ that is numerically
31 integrated to transport a simple source toward the data distribution, and that field has repeatedly been
32 leveraged to solve inverse problems without retraining [14–20].

33 This exposes a common limitation of standard FM priors. Because $v_\theta(x, t)$ represents only *instanta-*
34 *neous* motion, predicting a distant clean state requires either composing many solver steps or relying
35 on coarse extrapolation, which can degrade over long horizons. MeanFlow changes the interface:

rather than an instantaneous velocity, it learns average motion over a finite interval, enabling direct transport between widely separated times through a learned map $T_\theta(x_t; t \rightarrow t')$ [21–23]. Such maps have been exploited for test-time reward guidance [24, 25] and trained for conditional reconstruction [26, 27]; we ask a complementary question: *how does an off-the-shelf learned finite-interval map change training-free inverse-problem control?*

Our central claim is that these maps are useful not only for fast sampling but as *computational primitives for inference-time control*, since a learned map collapses a terminal look-ahead into one transition and supports algorithms that optimize against a predicted outcome without committing to an entire trajectory. We instantiate source-state optimization following D-Flow [17], PnP-Flow [18], MPC-RHC and MPC- Δt [20] on MeanFlow priors, and introduce *Receding-Horizon State Optimization* (RHSO), which optimizes the current state through its predicted clean endpoint, executes one interval, and replans from the state reached. Because a MeanFlow prior predicts any horizon at fixed cost, *where* to optimize becomes a design choice independent of how far ahead the solver looks.

Our contributions are: (i) establishing MeanFlow as a finite-interval controllable prior for training-free inverse problems and instantiating four representative solver structures on this interface without retraining; (ii) introducing RHSO and analyzing how intermediate-state intervention changes the prior-induced control geometry, with ablations isolating the benefit of repeated re-optimization; and (iii) providing a matched evaluation across five restoration tasks, showing that the pMF MeanFlow system is competitive with JiT overall and stronger in the few-step regime, and that RHSO is the strongest solver we evaluate on the MeanFlow prior.

2 Background

2.1 Flow Matching and finite-interval transport

Let $v_\theta : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$ be a learned velocity field inducing a probability path from a source distribution at $t = 0$ to the data distribution at $t = 1$. For $x_0 \sim p_{\text{source}}$ and $x_1 \sim p_{\text{data}}$, paired independently or by minibatch optimal transport [8], the linear conditional path $x_t = (1 - t)x_0 + tx_1$ has conditional velocity $\dot{x}_t = x_1 - x_0$, and Flow Matching regresses $\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, x_0, x_1} \|v_\theta(x_t, t) - (x_1 - x_0)\|_2^2$ without simulating the generative ODE during training [7]. Generation then solves

$$\frac{dx_t}{dt} = v_\theta(x_t, t), \quad x_0 \sim p_{\text{source}}, \quad (2)$$

with a numerical solver; we write $\Psi_{t \rightarrow t'}$ for exact transport, which a standard FM model approximates by composing solver steps. Denoising parameterizations may instead predict clean data, as in the pixel-space x -prediction of Li and He [28], which on the linear path induces $v_\theta(x_t, t) = [D_\theta(x_t, t) - x_t]/(1 - t)$. What matters is not the output parameterization but that conventional flow inference exposes *local* dynamics whose finite-time transport is obtained numerically.

MeanFlow instead learns the average velocity over a finite interval, parameterized directly by a network $u_\theta(x_t; t, t')$ [21]. Along a trajectory of equation (2),

$$u(x_t; t, t') = \frac{1}{t' - t} \int_t^{t'} v(x_\tau, \tau) d\tau, \quad x_{t'} = x_t + (t' - t) u(x_t; t, t'). \quad (3)$$

Differentiating equation (3) with respect to t at fixed t' yields the *MeanFlow identity* $u(x_t; t, t') - (t' - t)\dot{u}(x_t; t, t') = v(x_t, t)$, which links the average- and instantaneous-velocity fields through one Jacobian-vector product rather than a simulated trajectory and makes the objective trainable without simulation [22]. The learned field yields the finite-interval map

$$T_\theta(x_t; t \rightarrow t') = x_t + (t' - t) u_\theta(x_t; t, t'), \quad \lim_{t' \rightarrow t^+} u(x_t; t, t') = v(x_t, t). \quad (4)$$

The distinction is simple but consequential: $v_\theta(x_t, t)$ answers *where the state should move locally*, whereas $T_\theta(x_t; t \rightarrow t')$ answers *where it should be at the requested time t'* ; in particular $T_\theta(x_t; t \rightarrow 1)$ predicts the clean endpoint directly.

78 2.2 Training-free flow solvers for inverse problems

79 For equation (1), let

$$\Phi(x; y) = \frac{1}{2m} \|A(x) - y\|_2^2 \quad (5)$$

80 denote a representative fidelity objective over m measurements; exact normalizations are in Ap-
81 pendix C.

82 D-Flow optimizes the generative starting state and simulates the full trajectory from that state through
83 a terminal solve [17]. On a discretization $t = t_0 < \dots < t_K = 1$ with numerical updates F_j ,

$$x_t^* = \arg \min_{x_t} \Phi(F_{K-1} \circ \dots \circ F_0(x_t); y), \quad \nabla_{x_t} \Phi = J_{F_0}^\top \dots J_{F_{K-1}}^\top \nabla_{x_1} \Phi, \quad (6)$$

84 so every optimizer step backpropagates through the complete numerical trajectory. MPC-Flow
85 instead casts guidance as control, repeatedly planning over a finite horizon, executing one interval
86 and replanning [20, 29]; its best instantiation MPC- Δt , like PnP-Flow [18], relies on the one-step
87 clean-state estimate

$$D_t(x_t) = x_t + (1 - t)v_\theta(x_t, t), \quad (7)$$

88 which, for the linear path, is the optimal conditional-mean predictor $\mathbb{E}[x_1 \mid x_t]$ [9]. PnP-Flow
89 treats D_t as a denoiser inside an alternating fidelity-gradient, reprojection and denoising procedure
90 without backpropagating through the generative model [30, 18], while MPC- Δt uses it to define
91 a terminal value $\widehat{V}(t, x_t) = \Phi(D_t(x_t); y)$. Although optimal as a one-shot predictor, D_t need not
92 match the clean state reached by integrating the marginal flow; as a single Euler extrapolation, it
93 can incur substantial error over long jumps when the trajectory is curved [31]. Neither method
94 depends on any one prediction being exact, since both correct repeatedly. Diffusion-prior methods
95 span score-based, distilled and consistency models and enforce data consistency through projection,
96 guidance or posterior optimization [32–40]; our empirical study deliberately focuses on flow priors
97 and their deterministic transport.

98 Each method therefore reduces to a policy for repeatedly asking what a current state will become,
99 differing mainly in how that question is answered and how the answer is used.

100 3 MeanFlow control for inverse problems

101 3.1 A finite-interval solver interface

102 The solvers of section 2.2 share one primitive: a prediction of a future generative state, written
103 generically as $\mathcal{G}_\theta(x_t; t \rightarrow t')$ and, for $t' = 1$, as a prediction of the clean reconstruction. Standard
104 FM realizes $\mathcal{G}_\theta$ by composing solver steps; a MeanFlow prior realizes it with the single learned
105 evaluation of equation (4). Substituting one for the other instantiates each solver on a pretrained
106 MeanFlow prior, outer logic preserved and the model never fine-tuned: PnP-Flow uses $T_\theta(\cdot; t \rightarrow 1)$
107 as the prior projection in place of equation (7), D-Flow differentiates that single transport rather
108 than the composition in equation (6), and the MPC variants use T_θ for rollouts and terminal values
109 (Section A). Two properties matter below: a terminal prediction costs one model call at any horizon,
110 and its gradient is one backpropagation whose depth does not grow with the remaining trajectory.

111 3.2 Receding-horizon state optimization

112 Those two properties make a specific algorithm natural: optimize the current state through its
113 *predicted* clean reconstruction, commit one interval, and replan. We call this RHSO (Algorithm 1).
114 Unlike MPC-RHC, which optimizes explicit controls over a future rollout, RHSO directly optimizes
115 the current generative state; unlike D-Flow it repeats this at every stage, and unlike PnP-Flow it
116 differentiates the prior [20, 17, 18].

117 Let N be the number of receding-horizon stages, and let

$$0 = t_0 < t_1 < \dots < t_N = 1, \quad t_k = \left(\frac{k}{N}\right)^\beta, \quad k = 0, \dots, N, \quad (8)$$

118 be the stage grid for the pure-noise initialization used throughout, so N is also the number of
119 committed generative intervals. The uniform schedule $\beta = 1$ is the task-agnostic default in every

headline run; $\beta < 1$ concentrates stages toward the data endpoint and $\beta > 1$ toward the noise endpoint, with sensitivity analyzed in Section I. Write P_k for the terminal predictor available at t_k and E_k for the executed interval; both are supplied by the prior:

$$\text{MeanFlow: } P_k(x) = T_\theta(x; t_k \rightarrow 1), \quad E_k(x) = T_\theta(x; t_k \rightarrow t_{k+1}), \quad (9)$$

$$\text{FM, direct endpoint: } P_k(x) = D_{t_k}(x), \quad E_k(x) = \text{one numerical interval}, \quad (10)$$

$$\text{FM, integrated suffix: } P_k(x) = \text{numerical solve } t_k \rightarrow 1, \quad E_k(x) = \text{one numerical interval}. \quad (11)$$

Given a differentiable predictor P_k , the stage optimization is prior-agnostic. The direct FM planner is deliberately shallow and so computationally close to MeanFlow, whereas the integrated suffix is more faithful to the standard-flow trajectory but differentiates through every remaining interval as in equation (6); section 4.4 uses that contrast as a depth-matched control. Saving the pre-optimization state $s = x_{t_k}$, the stage objective is

$$x_{t_k}^* \approx \arg \min_x \left[\Phi(P_k(x); y) + \frac{\mu}{2d} \|x - s\|_2^2 \right], \quad (12)$$

solved only approximately, after which we commit $x_{t_{k+1}} = E_k(x_{t_k}^*)$ and repeat. The anchor weight μ penalizes departure from the state the prior would have produced; the headline benchmark sets $\mu = 0$ and section 4.5 intervenes on it directly.

Relation to one-shot state optimization. With $N = 1$, RHSO reduces to source-state optimization through a terminal predictor, the structure of D-Flow [17]; its distinguishing operation is the interleaving of optimization and execution, and since increasing N changes both where optimization is applied and how deep the predicted trajectory is, section 4.4 separates the two. MeanFlow does not remove optimization: it makes each terminal prediction a shallow learned operation whose depth does not grow with the remaining horizon, whereas standard FM must either approximate that horizon by one extrapolation or integrate it and differentiate through the result. Exact evaluation and backpropagation counts appear with the experiments.

3.3 Why intervention placement matters

Optimizing a state to improve a reconstruction is not unconstrained optimization of the image: the variable is a state x_t , the objective is evaluated on the endpoint, and the two are joined only by the remaining generative map, so $\delta x_1 \approx J_t \delta x_t$ with $J_t = D_x \Psi_{t \rightarrow 1}(x_t)$, and the fidelity gradient reaches the state as the pullback $J_t^\top \nabla_{x_1} \Phi$. Which endpoint corrections are economical is therefore dictated by J_t , a property of the learned prior that changes along the trajectory. We make this concrete in a Gaussian model, taking $p_{\text{data}} = \mathcal{N}(0, \Sigma)$ with $\Sigma e_i = s_i^2 e_i$ under the linear path of section 2.1. There $J_t e_i = g_i(t) e_i$ with gain $g_i(t) = s_i / \sqrt{t^2 s_i^2 + (1-t)^2}$, so $g_i(0) = s_i$ and $g_i(1) = 1$; derivations are in Section B and everything below is exact in that model only.

Source-state optimization inherits the spectral geometry of the prior. To probe the geometry the map induces, measure *intervention effort* by the squared Euclidean state displacement required to realize a prescribed endpoint correction $\delta = \sum_i \delta_i e_i$. From a single stage at t this is $\mathcal{C}_t(\delta) = \sum_i \delta_i^2 / g_i(t)^2$, so at the source

$$\mathcal{C}_0(\delta) = \sum_i \frac{\delta_i^2}{s_i^2} = \delta^\top \Sigma^{-1} \delta =: \|\delta\|_{\Sigma^{-1}}^2. \quad (13)$$

Algorithm 1 Receding-Horizon State Optimization (RHSO)

Require: observation y , fidelity Φ , stages N , schedule exponent β , anchor weight μ

- 1: $t_k \leftarrow (k/N)^\beta$ for $k = 0, \dots, N$ ▷ stage grid of equation (8); $t_0 = 0, t_N = 1$
 - 2: $x_{t_0} \leftarrow$ sample from the source distribution
 - 3: **for** $k = 0, \dots, N - 1$ **do**
 - 4: $s \leftarrow x_{t_k}$ ▷ save the state the prior actually reached
 - 5: $x_{t_k}^* \approx \arg \min_x [\Phi(P_k(x); y) + \frac{\mu}{2d} \|x - s\|_2^2]$ ▷ optimize through the predicted endpoint
 - 6: $x_{t_{k+1}} \leftarrow E_k(x_{t_k}^*)$ ▷ commit a single generative interval
 - 7: **end for**
 - 8: **return** x_{t_N}
-

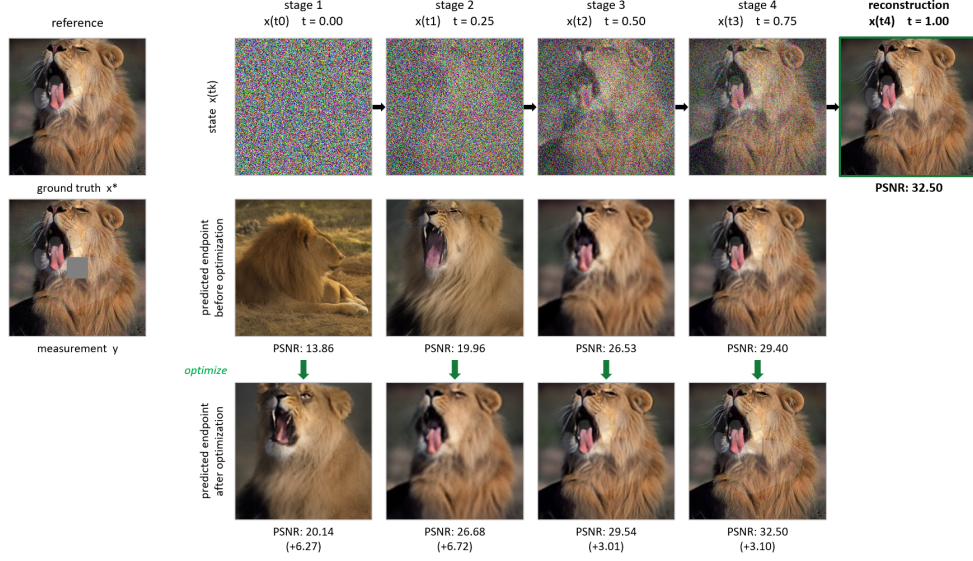


Figure 1: **What RHSO does over one trajectory** (MeanFlow prior, box inpainting, $N = 4$, $\beta = 1$, so $t_k \in \{0, .25, .5, .75\}$). *Top*: the state after each committed interval. *Below*: the terminal prediction entering each stage, $P_k(x_{t_k})$, and the one leaving it, $P_k(x_{t_k}^*)$. Every stage improves its own readout, by 6.27 and 6.72 dB at the two noisiest states, so each committed interval poses a new problem rather than refining one plan. Early predictions are poor because the state is nearly pure noise; what matters is the per-stage improvement.

152 Its anisotropy is $\kappa(\Sigma) = \lambda_{\max}(\Sigma)/\lambda_{\min}(\Sigma) = s_{\max}^2/s_{\min}^2$: for equal-magnitude corrections $\delta = e_i$
 153 the most expensive costs $\kappa(\Sigma)$ times the least, so directions in which the prior varies little are exactly
 154 the expensive ones to move. Note $\mathcal{C}_t$ is an analytical control-effort metric, not a term any solver
 155 minimizes—with $\mu = 0$ nothing in equation (12) penalizes displacement. The prior-shaped geometry
 156 is a property of the landscape, which can behave as implicit regularization under a finite budget.

157 Distributed optimization introduces independent perturbations propagated through different suffix
 158 geometries. Under the same effort, the optimal allocation over $\mathcal{T} = \{t_k\}$ costs

$$\mathcal{C}_{\mathcal{T}}^*(\delta) = \sum_i \frac{\delta_i^2}{\sum_k g_i(t_k)^2} = \delta^\top W_{\mathcal{T}}^{-1} \delta, \quad W_{\mathcal{T}} = \sum_k J_{t_k} J_{t_k}^\top, \quad (14)$$

159 which recovers equation (13) when $\mathcal{T} = \{0\}$, since $W_{\{0\}} = \Sigma$. The question is how the *shape* of
 160 $W_{\mathcal{T}}$ differs from Σ .

161 **Proposition 1** (Later stages reduce spectral contrast). *Assume the prior is anisotropic, $s_{\max} > s_{\min}$,
 162 and let $\chi(\mathcal{T})$ be the ratio of the largest to the smallest diagonal entry of $W_{\mathcal{T}}$ in the covariance
 163 eigenbasis $\{e_i\}$, so $\chi(\{0\}) = \kappa(\Sigma)$. If $t' > \max \mathcal{T}$, then $\chi(\mathcal{T} \cup \{t'\}) < \chi(\mathcal{T})$.*

164 Here χ is a ratio of sums lying between the per-stage contrasts $g_{\max}(t_k)^2/g_{\min}(t_k)^2$, which decrease
 165 in t , so a later, less anisotropic stage pulls the aggregate down. On a dense uniform grid this saturates
 166 at $\mathcal{C}^* \sim \frac{2}{\pi N} \|\delta\|_{\Sigma^{-1/2}}^2$, a contrast of $\sqrt{\kappa(\Sigma)}$ rather than $\kappa(\Sigma)$ (Section B); the $1/N$ prefactor is
 167 generic to splitting a displacement across quadratically penalized controls, so the prior-related content
 168 is the change of *shape* from Σ^{-1} toward $\Sigma^{-1/2}$. Prediction depth and intervention placement are
 169 therefore distinct: an exact source-to-endpoint predictor still exposes only the source geometry,
 170 whereas repeated optimization accesses several intermediate ones, which the analysis suggests
 171 tempers it. While this analysis is exact only for the idealized Gaussian setting, it motivates distributed
 172 state optimization across multiple timesteps: intervening at different stages exposes the optimizer to
 173 distinct prior-induced geometries, thus motivating RHSO and, more broadly, repeated intervention
 174 in MPC-style flow control. We test whether this mechanism carries over to learned nonlinear priors
 175 below.

4 Experiments

Setup. Following the 100-image evaluation protocol used in previous flow-based inverse-problem benchmarks [18, 20], we evaluate Gaussian denoising, Gaussian deblurring, $2\times$ super-resolution, random inpainting and box inpainting on 100 ImageNet-1k validation images from distinct classes [41], requiring known source-class conditioning for both priors. The priors are pixel-space pMF-L/16 (CFG 7) [23] and JiT-B/16 (CFG 3) [28]; because they differ in capacity, training and guidance, cross-prior results are system-level. Equation (12) uses M Adam steps warm-started at s [42], so $B = NM$. Unless ablated, headline runs use $N = 4$, $M = 40$, pure prior noise, $\beta = 1$, $\mu = 0$, and configurations frozen after equal tuning effort. We report PSNR, SSIM [43], LPIPS [44], and measurement RMSE; protocol, hyperparameters and full tables are in Sections C, D and J.

4.1 MeanFlow is a strong controllable prior

Table 1 reports the main benchmark. All four solver structures run on the pretrained MeanFlow prior without retraining and without change to their outer logic, so the finite-interval interface carries existing designs rather than one bespoke algorithm.

Table 1: **Unified 100-image reconstruction benchmark.** PSNR $\uparrow$ /SSIM $\uparrow$ /LPIPS $\downarrow$; bold marks the best value per prior block and task. “D-Flow (1-step)” denotes our one-terminal-prediction instantiation of the D-Flow source-state objective; it is not a reproduction of the original multi-step D-Flow solver or its hyperparameters [17]; row two of table 2 is depth-matched.

Prior	Method	Denoising	Deblurring	Super-res.	Rand. inpaint.	Box inpaint.
pMF	PnP-Flow	27.99/.752/.127	28.86/.823/.175	26.09/.769/.129	26.04/.767/.127	28.52/.870/.065
	D-Flow (1-step)	25.19/.671/.285	23.32/.605/.377	24.80/.673/.265	24.91/.673/.263	24.20/.683/.285
	MPC-RHC	29.02/.808/.110	28.91/.796/.145	24.96/.689/.233	24.82/.678/.237	25.79/.842/.089
	MPC- Δt	28.30/.737/.151	29.75/.836/.122	27.07/.785/.108	28.03/.831/.091	29.58/.889/.051
	RHSO	28.47/.783/.150	29.84/.858/.105	27.78/.819/.098	28.17/.840/.079	28.26/.893/.068
JiT	PnP-Flow	28.92/.795/.099	29.10/.844/.166	27.50/.793/.138	27.93/.825/.089	30.81/.874/.058
	D-Flow (1-step)	19.10/.363/.656	18.99/.355/.659	19.07/.363/.655	19.09/.363/.653	18.92/.360/.657
	MPC-RHC	28.99/.798/.119	29.15/.801/.165	27.07/.768/.141	26.82/.759/.146	27.57/.844/.082
	MPC- Δt	28.05/.727/.176	30.02/.843/.123	27.78/.801/.090	28.75/.851/.073	30.37/.891/.049
	RHSO direct	27.79/.734/.198	29.26/.845/.144	27.71/.816/.102	28.20/.835/.094	29.39/.891/.074
	RHSO suffix	27.43/.698/.267	30.15/.870/.107	27.93/.817/.104	28.58/.847/.075	29.87/.906/.051

At this operating point the priors are close and neither dominates: the best pMF configuration averages 28.50 dB/.839/.100 and the best JiT configuration 28.99/.823/.102, with rankings flipping by solver. The D-Flow (1-step) stress test averages 19.03 dB for JiT versus 24.48 dB for pMF; it probes direct terminal prediction and should not be read as full D-Flow performance.

4.2 RHSO is strongest on the MeanFlow prior

On the MeanFlow prior, RHSO has the better value in 52/60 task–metric cells (descriptive counts over correlated metrics): 15/15 against D-Flow (1-step), 13/15 against MPC- Δt , and 12/15 against each of PnP-Flow and MPC-RHC. It is best in 10/15 cells, with exceptions concentrated in denoising and box inpainting. Across table 1, pMF RHSO has the best five-task mean SSIM (.839) and LPIPS (.100), while mean PSNR (28.50 dB) is within 0.49 dB of the best. It uses 164/160 Eval./BP at 14.28 s/image; the depth-matched standard-flow variant needs 647/640 and 56.65 s/image for 0.29 dB more PSNR but worse SSIM/LPIPS. Note that this is not an iso-compute ranking, for PnP-Flow is much cheaper than the other methods (Section K).

4.3 The few-step regime

The pMF–JiT gap should be sharpest when few generative intervals are available, since that is when the JiT planner must extrapolate far or integrate a long suffix. We sweep the stage count N under two budget controls: *fixed optimizer work* $B = NM = 160$, isolating the trajectory budget, and *fixed per-stage work* $M = 40$. Figure 3 shows five-task means; Section F gives task-specific curves.

At $N \leq 2$ pMF has better values in all 60 task–metric cells across both protocols: by 3.37/1.17 dB at $N = 1/2$ with fixed optimizer work and 1.02/1.69 dB with fixed per-stage work; mean LPIPS at

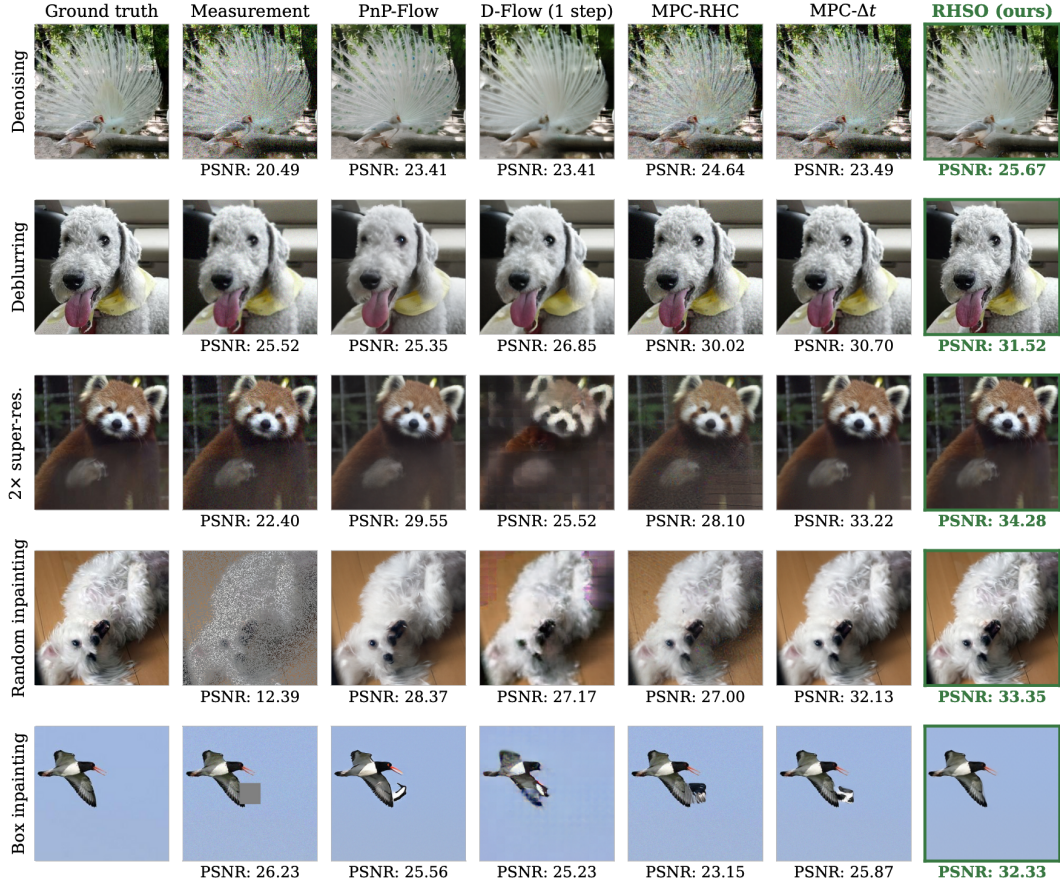


Figure 2: **Qualitative examples from the MeanFlow benchmark** (per-image PSNR below each panel). PnP-Flow and D-Flow (1-step) over-smooth texture; MPC variants leave residual denoising noise; on the shown box-inpainting example, RHSO reconstructs a coherent missing wing. These fixed examples illustrate behavior rather than establish aggregate superiority; table 1 reports aggregate metrics.

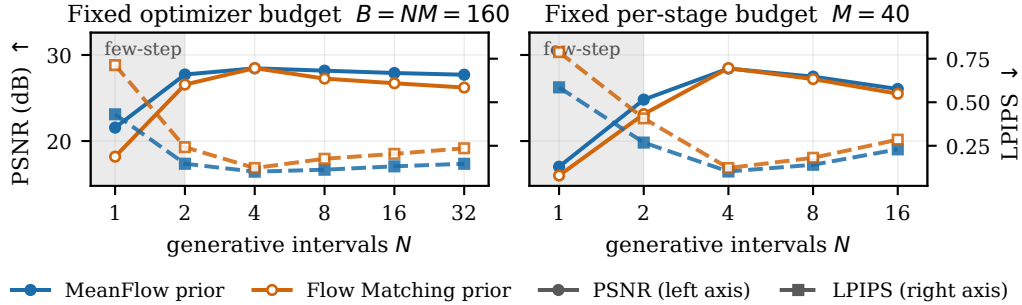


Figure 3: **pMF is competitive throughout and clearly stronger in the few-step regime**. Five-task means under two budget controls. At $N \leq 2$ pMF has better values in all 60 task-metric cells across both panels; at $N = 4$ the priors are tied in PSNR while MeanFlow keeps a perceptual margin; beyond that both degrade, MeanFlow more slowly.

210 $N = 1$ is .431 versus .713. At $N = 4$, where the headline benchmark sits, the priors are tied in mean
 211 PSNR (-0.02 dB) while pMF is better on all five tasks in SSIM and LPIPS. Across all stage counts
 212 it has better values in 86/90 cells at fixed B and 71/75 at fixed M , including all 55 SSIM and all
 213 55 LPIPS cells; the eight exceptions are PSNR, six on box inpainting. Together with the D-Flow
 214 (1-step) contrast above, this shows that the finite-interval interface is particularly advantageous when
 215 the trajectory budget is small.

216 4.4 Placement versus prediction depth

217 Moving from $N = 1$ to $N = 4$ changes two things at once: where optimization is applied, and how
 218 deep the predicted trajectory is. To separate them we add a one-shot standard-flow control keeping
 219 all 160 updates at the source but differentiating each through the same four-interval Heun suffix,
 220 matching RHSO’s trajectory depth while withholding its extra intervention points.

Table 2: **Separating optimization placement from numerical suffix depth** (standard-flow prior, five-task averages, Eval./BP per image). Matching prediction depth recovers little; redistributing the same 160 updates recovers most.

Variant	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Eval./BP $\downarrow$
One-shot, direct endpoint	18.18	.375	.713	161/160
One-shot, 4-step integrated suffix	20.84	.500	.551	1127/1120
RHSO, direct endpoint	28.47	.824	.122	167/160
RHSO, integrated suffix	28.79	.828	.121	647/640

221 Deepening the one-shot predictor adds $+2.67$ dB across tasks ($+2.57$ to $+2.75$ dB; all paired intervals
 222 positive; Section E), so coarse endpoint prediction partly explains the weak source-only baseline.
 223 Redistributing the same 160 updates across four stages adds $+7.63$ dB; with replanning, a full suffix
 224 adds only $+0.32$ dB at four times the model-side cost. Thus *re-optimizing intermediate states*,
 225 not endpoint accuracy, is dominant. This is qualitatively consistent with the idealized analysis in
 226 section 3.3: deeper one-shot prediction evaluates the same map J_0 , whereas distributed optimization
 227 introduces independent controls through different suffix geometries (equation (14)). It also explains
 228 why RHSO tolerates a shallow planner: each execution replans from the reached state (figure 1).

229 4.5 When additional control becomes excessive

230 Distributing intervention tempers the prior-shaped constraint of section 3.3, but with $\mu = 0$ fidelity
 231 alone acts on the reconstruction, so greater authority need not improve image quality. At fixed
 232 $M = 40$, $N = 4 \rightarrow 16$ improves RMSE in all ten prior–task pairs but degrades PSNR on all five
 233 tasks for both priors; for pMF super-resolution, RMSE falls $.052 \rightarrow .041$ while PSNR drops 4.04 dB
 234 and LPIPS rises $.100 \rightarrow .321$. Fixed $B = 160$ reproduces the interior optimum, ruling out optimizer
 235 starvation. The useful budget also shrinks as the observation degrades; an anchor $\mu = 1$ at $N = 8$
 236 recovers 2.12 dB on the standard-flow prior and 0.96 dB on MeanFlow at worse RMSE—consistent
 237 with the implicit-control account, not a controlled test. Full curves and the noise, anchor, schedule
 238 and initialization studies are in Sections F to I.

239 5 Conclusion and further work

240 Pretrained MeanFlow models provide controllable priors for training-free inverse problems, enabling
 241 direct terminal predictions at any horizon. Four representative solver structures transfer without
 242 retraining; across five ImageNet restoration tasks, pMF is competitive with JiT overall and stronger
 243 at small trajectory budgets. The interface also makes RHSO natural: it is the strongest MeanFlow
 244 solver we evaluate, while a matched control attributes its advantage to re-optimizing intermediate
 245 states. However, RHSO remains expensive (NM backpropagations), the priors are distinct pretrained
 246 systems, and our Gaussian analysis is idealized and does not establish causality (Section L). These
 247 limitations motivate lower-overhead variants and further tests of the mechanism in learned nonlinear
 248 priors. Beyond inverse problems, equation (5) is only one possible constraint: equation (12) supports
 249 other differentiable objectives—including semantic, counterfactual, safety, and constrained generation
 250 [45–49]—and multimodal, language-conditioned MeanFlow priors could broaden such guidance.

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A Additional methodological details

A.1 Solver adaptations

The main paper writes the generative transition available to a solver as $\mathcal{G}_\theta(x_t; t \rightarrow t')$. Standard FM obtains this transition by numerically integrating the instantaneous velocity field; MeanFlow evaluates the learned map in equation (4). This substitution leaves the outer inverse-solver logic unchanged.

D-Flow (1-step) source-state optimization. Following the source-state objective of Ben-Hamu et al. [17], our D-Flow (1-step) variant optimizes a single source or intermediate state through one terminal predictor,

$$x_{t_0}^* = \arg \min_x \Phi(\mathcal{G}_\theta(x; t_0 \rightarrow 1); y). \quad (15)$$

If one terminal solve uses E_D generative-model evaluations and the optimizer takes M updates, the model-side cost is $E_D(M + 1)$ evaluations and $E_D M$ backpropagations. Direct MeanFlow uses $E_D = 1$. This is a one-terminal-prediction adaptation of the D-Flow objective, not a reproduction of the original multi-step D-Flow solver or its reported hyperparameters. For JiT, the one-step variant replaces the full terminal solve by equation (7); the depth-matched JiT suffix control integrates four Heun intervals before evaluating the terminal loss.

PnP-Flow. Following Martin et al. [18], PnP-Flow alternates a fidelity update and a time-dependent prior projection. The standard-flow prior is $D_t(x) = x + (1 - t)v_\theta(x, t)$, while the MeanFlow prior is $T_\theta(x; t \rightarrow 1)$. The selected implementation uses one prior realization per iteration and no generative-model backpropagation.

MPC-Flow. MPC-RHC and MPC- Δt optimize explicit controls rather than the generative state [20], following the receding-horizon principle standard in model-predictive control [29]. MeanFlow replaces each nominal rollout transition by T_θ . For MPC-RHC, a cached first transition means that a $K = 1$ horizon requires no model backpropagation through the optimized objective; larger horizons differentiate through the remaining transitions. MPC- Δt caches the next short transition and evaluates a terminal value from the following state. Exact selected horizons and control budgets are listed in Appendix D.

A.2 RHSO implementation and costs

At every outer stage, the optimized state is initialized from the state actually reached and a fresh Adam state is created. Optimizer momentum and computation graphs are not propagated between stages. The anchor in equation (12) uses a stop-gradient copy of the reached state and remains fixed during all M inner updates.

For pMF, N stages and M inner updates require

$$C_{\text{eval}}^{\text{pMF}} = N(M + 1), \quad C_{\text{bp}}^{\text{pMF}} = NM. \quad (16)$$

For the direct JiT planner, there are NM one-step terminal predictions plus the executed Heun/Euler intervals. At $N = 4, M = 40$, this gives 167/160 Eval./BP. For full-suffix JiT, a suffix of j intervals uses $2j - 1$ evaluations, so

$$C_{\text{plan}}^{\text{suffix}} = M \sum_{j=1}^N (2j - 1) = MN^2, \quad C_{\text{exec}}^{\text{suffix}} = 2N - 1. \quad (17)$$

The selected $N = 4, M = 40$ configuration therefore uses 647/640 Eval./BP. The four-step one-shot suffix uses the same seven-evaluation terminal graph for every one of 160 source-state updates, giving 1127/1120 Eval./BP after the final generation pass.

Collecting these at the headline setting $N = 4, M = 40$: every prior performs the same $NM = 160$ optimizer updates, and only the *depth* of each terminal gradient differs. The MeanFlow prior uses 164 evaluations and 160 backpropagations per image and the direct standard-flow planner 167/160, whereas the full-suffix planner differentiates the remaining 4, 3, 2, 1 intervals and uses 647/640. This is the concrete sense in which a finite-interval map makes each terminal gradient shallow and independent of the remaining horizon: it does not remove the optimization, only the growth of its cost with the look-ahead.

B Theory details and proofs

B.1 Gaussian terminal map and gain profile

Assume $x_1 \sim \mathcal{N}(0, \Sigma)$ and the CondOT path

$$x_t = tx_1 + (1 - t)x_0, \quad x_0 \sim \mathcal{N}(0, I), \quad (18)$$

with x_0, x_1 independent, so that $x_t \sim \mathcal{N}(0, C_t)$ with $C_t = t^2\Sigma + (1 - t)^2I$. We derive $\Psi_{t \rightarrow 1}$ from the flow-matching dynamics rather than positing a covariance-matching map. The optimal marginal velocity is the conditional expectation $u_t(x) = \mathbb{E}[x_1 - x_0 \mid x_t = x]$, and $(x_t, x_1 - x_0)$ is jointly Gaussian with zero mean and $\text{Cov}(x_1 - x_0, x_t) = t\Sigma - (1 - t)I$, so

$$u_t(x) = M_t x, \quad M_t = (t\Sigma - (1 - t)I) C_t^{-1}. \quad (19)$$

The velocity is linear, hence so is the flow. All of Σ, C_t and M_t share the eigenbasis of Σ and therefore commute, so the generative ODE decouples into scalar equations. In the eigenmode with Σ -eigenvalue s^2 , writing $c_t = t^2s^2 + (1 - t)^2$ for the corresponding entry of C_t ,

$$M_t = \frac{ts^2 - (1 - t)}{c_t} = \frac{\frac{1}{2}\dot{c}_t}{c_t} = \frac{d}{dt} \left[\frac{1}{2} \log c_t \right], \quad (20)$$

438 since $\dot{c}_t = 2ts^2 - 2(1-t)$. Integrating from t to 1 and using $c_1 = s^2$ gives the scalar flow factor
 439 $\exp[\frac{1}{2} \log(s^2/c_t)] = s/\sqrt{c_t}$. Collecting eigenmodes,

$$\Psi_{t \rightarrow 1}(x) = \Sigma^{1/2} C_t^{-1/2} x, \quad (21)$$

440 so this map is the exact terminal transport of the Gaussian flow-matching dynamics, not merely one
 441 covariance-matching choice. Differentiating it gives $J_t = \Sigma^{1/2}(t^2 \Sigma + (1-t)^2 I)^{-1/2}$, which is
 442 diagonal in $\{e_i\}$ with the gain profile $g_i(t)$ quoted in Section 3.3: $g_i(t)$ is the endpoint displacement
 443 produced per unit state displacement along e_i . In a covariance eigenmode with standard deviation
 444 $s > 0$,

$$g(t) = \frac{s}{\sqrt{(1+s^2)t^2 - 2t + 1}}. \quad (22)$$

445 The denominator is a strictly convex quadratic with derivative $2(1+s^2)t - 2$, hence g is unimodal
 446 with unique maximum $g(t^*) = \sqrt{1+s^2}$ at $t^* = 1/(1+s^2)$. The endpoints satisfy $g(0) = s$ and
 447 $g(1) = 1$, which is what makes the source geometry of equation (13) exactly Σ^{-1} .

448 For $s_{\max} > s_{\min}$, the squared anisotropy is

$$\kappa(t)^2 = \frac{s_{\max}^2 t^2 s_{\min}^2 + (1-t)^2}{s_{\min}^2 t^2 s_{\max}^2 + (1-t)^2}. \quad (23)$$

449 Writing $r = t^2/(1-t)^2$, the nonconstant factor is $(rs_{\min}^2 + 1)/(rs_{\max}^2 + 1)$, whose derivative in r is
 450 strictly negative. Thus $\kappa(t)$ decreases to one. Note that the largest individual gain need not decrease
 451 monotonically; it inherits the unimodal profile above.

452 B.2 Time-average identity

453 Completing the square yields

$$\int_0^1 g(t)^2 dt = \int_0^1 \frac{s^2}{(1+s^2)t^2 - 2t + 1} dt \quad (24)$$

$$= s [\arctan u]_{u=-1/s}^{u=s} = \frac{\pi}{2} s, \quad (25)$$

454 where the final equality uses $\arctan(s) + \arctan(1/s) = \pi/2$ for $s > 0$.

455 B.3 Idealized stage allocation and the control Gramian

456 Let stage k contribute endpoint displacement $\delta_i^{(k)} = g_i(t_k) \Delta_i^{(k)}$, and assume these endpoint effects
 457 compose additively, so that a prescribed total correction satisfies $\delta = \sum_k \delta^{(k)}$. This idealization is
 458 what yields the Gramian form of equation (14). We solve

$$\min_{\sum_k \delta^{(k)} = \delta} \sum_{k,i} \frac{(\delta_i^{(k)})^2}{g_i(t_k)^2}. \quad (26)$$

459 The problem separates over i . For each direction, a Lagrange multiplier gives

$$\delta_i^{(k)} = \delta_i \frac{g_i(t_k)^2}{\sum_j g_i(t_j)^2}, \quad (27)$$

460 with optimum

$$\mathcal{C}_{\mathcal{T}}^*(\delta) = \sum_i \frac{\delta_i^2}{\sum_k g_i(t_k)^2} = \delta^\top W_{\mathcal{T}}^{-1} \delta, \quad (28)$$

461 the last equality because J_{t_k} is diagonal in $\{e_i\}$ with entries $g_i(t_k)$, so $W_{\mathcal{T}} = \sum_k J_{t_k} J_{t_k}^\top$ is diagonal
 462 with entries $\sum_k g_i(t_k)^2$. For $\mathcal{T} = \{0\}$ this is $W_{\{0\}} = \Sigma$ and recovers equation (13). This is an
 463 allocation result, not a statement that adding an RHSO stage cannot hurt: actual stages are sequential,
 464 later gains depend on earlier state updates, and each stage is optimized only for a finite number of
 465 steps.

466 B.4 Proof of Proposition 1

467 Throughout we assume the prior is anisotropic, $s_{\max} > s_{\min}$; if $s_{\max} = s_{\min}$ then $\chi \equiv 1$ for every
 468 stage set and no strict reduction is available. Because $g_i(t)$ is strictly increasing in s_i for every $t < 1$,
 469 the largest and smallest diagonal entries of $W_{\mathcal{T}}$ are attained at $s_{\max}$ and $s_{\min}$, so

$$\chi(\mathcal{T}) = \frac{\sum_k g_{\max}(t_k)^2}{\sum_k g_{\min}(t_k)^2}, \quad \chi(\{0\}) = \frac{s_{\max}^2}{s_{\min}^2} = \kappa(\Sigma). \quad (29)$$

470 For positive a_k, b_k the ratio $\sum_k a_k / \sum_k b_k$ lies between $\min_k a_k/b_k$ and $\max_k a_k/b_k$, so $\chi(\mathcal{T})$
 471 lies between the per-stage contrasts $\kappa(t_k)^2 = g_{\max}(t_k)^2/g_{\min}(t_k)^2$; in particular $\chi(\mathcal{T}) \geq$
 472 $\min_k \kappa(t_k)^2 = \kappa(\max \mathcal{T})^2$, using that $\kappa(t)$ is strictly decreasing. Adjoining $t' > \max \mathcal{T}$ gives
 473 $\kappa(t')^2 < \kappa(\max \mathcal{T})^2 \leq \chi(\mathcal{T})$, using strict monotonicity of $\kappa(t)$, which requires $s_{\max} > s_{\min}$. Since
 474 the median $\frac{A+a'}{B+b'}$ lies strictly between A/B and a'/b' whenever these differ, $\chi(\mathcal{T} \cup \{t'\}) < \chi(\mathcal{T})$.
 475 The statement concerns the shape of $W_{\mathcal{T}}$ only; it does not assert that additional stages improve recon-
 476 struction, and it says nothing about the $1/N$ scale factor, which is generic to splitting a displacement
 477 across quadratically penalized controls.

478 B.5 Uniform-grid asymptotics

479 For the left grid $t_k = k/N$, the function g_i^2 has total variation $1 + s_i^2$, so the bounded-variation
 480 Riemann estimate gives

$$\left| \frac{1}{N} \sum_{k=0}^{N-1} g_i(t_k)^2 - \frac{\pi}{2} s_i \right| \leq \frac{1 + s_i^2}{N}. \quad (30)$$

481 Whenever the relative error is uniformly small over the directions of interest,

$$\sum_k g_i(t_k)^2 = \frac{\pi N}{2} s_i (1 + o(1)), \quad (31)$$

482 which inserted into the allocation optimum gives the dense-grid asymptotic $\mathcal{C}^* \sim \frac{2}{\pi N} \|\delta\|_{\Sigma^{-1/2}}^2$
 483 quoted in Section 3.3. At $N = 1$, the exact cost is $\|\delta\|_{\Sigma^{-1}}^2$; in the dense-grid limit the spectral
 484 exponent is tempered to $\Sigma^{-1/2}$. No finite- N power law is implied.

485 B.6 A closed-loop correction bound

486 The allocation argument explains possible geometric benefits of multiple locations but does not model
 487 prediction error. A complementary model-agnostic bound captures the value of replanning from the
 488 reached state. Let P_k be the terminal predictor, E_k the executed short transition, and

$$\epsilon_k = \|P_k(x_{t_k}^*) - P_{k+1}(E_k(x_{t_k}^*))\|. \quad (32)$$

489 If Φ is L_{Φ} -Lipschitz in terminal image space and stage k lowers its own predicted terminal objective
 490 by $\Delta_k \geq 0$, then

$$V_{k+1} \leq V_k - \Delta_k + L_{\Phi} \epsilon_k. \quad (33)$$

491 If the optimized value obeys $V_k^+ \leq (1 - \theta_k) V_k$, iteration gives

$$V_N \leq \left[\prod_k (1 - \theta_k) \right] V_0 + L_{\Phi} \sum_k \left[\prod_{j>k} (1 - \theta_j) \right] \epsilon_k. \quad (34)$$

492 The bound formalizes only that later replanning reconditions the objective on the state actually
 493 reached. It does not assert that ϵ_k is small or that later optimization is intrinsically faster.

494 B.7 Toy Example

495 C Experimental protocol and reproducibility

496 C.1 Dataset, models, and pairing

497 We use 100 ImageNet-1k validation images at 256×256 , one from each of 100 distinct classes, and
 498 condition every reconstruction on the known source class. The exact image identifiers used in the

analyses are provided in the anonymized code repository as `data/image_ids.txt`, alongside the implementation and configurations needed to reproduce the reported experiments. The anonymized repository is available at <https://anonymous.4open.science/r/flow-review-2026-4221/>. The pMF prior is pMF-L/16 at training step 400320 with classifier-free-guidance scale 7. The JiT prior is JiT-B/16 with classifier-free-guidance scale 3. JiT network inference uses bfloat16 with float32 integration in the audited runs.

For a fixed source image and model, method identity does not enter the random seeds used to construct the inverse problem or initial generative noise. Paired comparisons therefore share source image, class, forward operator, measurement realization, inpainting mask, canonical start time, schedule, and initial Gaussian noise. Follow-up archives additionally record measurement IDs, guide IDs, and initial-state fingerprints, which were checked for parity before paired statistics were computed.

C.2 Forward operators

The five tasks use the following fixed degradation parameters:

- **Denoising:** $y = x^* + \eta$, $\eta \sim \mathcal{N}(0, .2^2 I)$.
- **Deblurring:** a 7×7 Gaussian kernel with blur standard deviation 1.0 and reflect padding, followed by Gaussian noise with standard deviation .05.
- **Super-resolution:** true $2 \times$ spatial downsampling with Gaussian measurement noise of standard deviation .05. Bicubic upsampling is used only as the optional image-space guide.
- **Random inpainting:** 70% missing pixels with Gaussian noise of standard deviation .01 on observed entries.
- **Box inpainting:** a 40×40 missing box with Gaussian noise of standard deviation .05 on observed entries.

The denoising guide is the observation, deblurring uses the observed blurred image, super-resolution uses bicubic upsampling, and inpainting uses zero filling. In headline experiments the canonical initialization time is zero, so the state starts from pure prior noise and the guide contributes no image content to the initial state.

C.3 Fidelity normalization and metrics

D-Flow (1-step), PnP-Flow, and RHSo use the half mean-squared residual per measured scalar,

$$\Phi(x; y) = \frac{1}{2m} \|A(x) - y\|_2^2, \quad (35)$$

with a batch sum so each image gradient is invariant to computational batch size. The inherited MPC implementations use their selected sum-normalized objective and control cost. We report PSNR, SSIM, LPIPS, and

$$\text{RMSE}_{\text{meas}} = \frac{\|A(\hat{x}) - y\|_2}{\sqrt{m}}. \quad (36)$$

For inpainting we also report observed-region and missing-region quantities when they clarify global metric behavior.

C.4 Statistical analysis

All principal reconstruction comparisons preserve image-level pairing. Paired 95% confidence intervals are nonparametric bootstrap intervals over the 100 paired image differences, using seed 20260903. We emphasize effect sizes and metric agreement rather than binary significance decisions. Measurement RMSE in several compact archives is stored only as an aggregate job-level diagnostic and is therefore reported descriptively rather than treated as 100 independent samples.

C.5 Compute environment and third-party assets

The fixed-budget, fixed-per-stage, controlled-noise, and state-anchor follow-ups were run on NVIDIA A100-SXM4-40GB GPUs. The schedule sweep was run on an NVIDIA H100 80GB HBM3 GPU.

The dedicated batch-one resource comparison in Appendix K uses an A100-SXM4-40GB. The result archives record accelerator identity, model provenance, seeds, and elapsed run metadata in addition to the resolved configurations.

All assets used in this work are pre-existing and are credited to their original sources. The evaluation images are ImageNet-1k validation images [41], accessed under the ImageNet terms of use, which grant researchers a non-exclusive right to use the images for non-commercial research and educational purposes. The generative priors are the pretrained checkpoints released with the official JiT [28] and pMF [23] implementations; both reference repositories are distributed under the MIT license, and we additionally comply with any terms accompanying the released checkpoint files themselves. We use these assets only for inference-time evaluation and do not fine-tune or otherwise modify the released weights. This manuscript source package redistributes no ImageNet images and no third-party pretrained checkpoints, and releases no new dataset or pretrained generative model.

D Frozen solver configurations

All headline configurations use canonical $t_0 = 0$, $\beta = 1$, and $\mu = 0$. Optimization-based methods use Adam. Calibration parameters may differ across model families, but both families received equal tuning effort and the same structural budget where the comparison requires it. Tables 3 and 4 list the final MeanFlow-prior configurations and tables 5 and 6 the standard-flow ones.

Selection proceeded in two stages. A coarse screening on 2 images narrowed each solver’s grid, after which a focused refinement on 8 images fixed the reported values. The refinement stage used a shared deterministic 8-image subset of the curated ImageNet validation pool, identical for both priors, so that configurations were compared on the same images. The JiT and pMF searches were matched in structural and computational effort, so neither prior received a larger or deeper grid than the other. Configurations were chosen after examining reconstruction-quality metrics during this matched refinement, including perceptual quality as measured by LPIPS rather than a single scalar criterion, and were then frozen before the 100-image evaluation reported throughout the paper.

Table 3: **Final pMF hyperparameters: denoising, deblurring, and super-resolution.**

Method	Denoising	Deblurring	Super-resolution
	$N = 50, \gamma_0 = 8 \times 10^5$	$N = 100, \gamma_0 = 1.2 \times 10^6$	$N = 20, \gamma_0 = 10^5$
PnP-Flow	$\alpha = .75$	$\alpha = .50$	$\alpha = .25$
D-Flow (1-step)	$M = 320, \text{lr} = .03$	$M = 320, \text{lr} = .03$	$M = 320, \text{lr} = .03$
	$N = 4, K = 2, C = 40$	$N = 4, K = 1, C = 40$	$N = 4, K = 3, C = 40$
MPC-RHC	$\lambda = 15, \text{lr} = .10$	$\lambda = 60, \text{lr} = .05$	$\lambda = 240, \text{lr} = .05$
	$N = 8, C = 40$	$N = 8, C = 40$	$N = 4, C = 20$
MPC- Δt	$\lambda = 30, \text{lr} = .10$	$\lambda = 540, \text{lr} = .10$	$\lambda = 360, \text{lr} = .10$
RHSO	$N = 4, M = 40, \text{lr} = .03$	$N = 4, M = 40, \text{lr} = .01$	$N = 4, M = 40, \text{lr} = .01$

Table 4: **Final pMF hyperparameters: inpainting.**

Method	Random inpainting	Box inpainting
	$N = 20, \gamma_0 = 10^5$	$N = 100, \gamma_0 = 4 \times 10^5$
PnP-Flow	$\alpha = .25$	$\alpha = .25$
D-Flow (1-step)	$M = 320, \text{lr} = .03$	$M = 320, \text{lr} = .03$
	$N = 4, K = 3, C = 40$	$N = 4, K = 1, C = 40$
MPC-RHC	$\lambda = 180, \text{lr} = .05$	$\lambda = 30, \text{lr} = .05$
	$N = 4, C = 20$	$N = 8, C = 40$
MPC- Δt	$\lambda = 540, \text{lr} = .15$	$\lambda = 180, \text{lr} = .10$
RHSO	$N = 4, M = 40, \text{lr} = .01$	$N = 4, M = 40, \text{lr} = .02$

All optimization-based pMF configurations use Adam; RHSO uses $\mu = 0$. The standard-flow configurations follow.

Table 5: **Final JiT hyperparameters: denoising, deblurring, and super-resolution.** Standard-flow trajectory integrations use Heun with the Euler final-step policy where applicable.

Method	Denoising	Deblurring	Super-resolution
PnP-Flow	$N = 50, \gamma_0 = 2 \times 10^5$ $\alpha = .50$	$N = 100, \gamma_0 = 1.2 \times 10^6$ $\alpha = .50$	$N = 20, \gamma_0 = 10^5$ $\alpha = .25$
D-Flow (1-step)	$M = 320, \text{lr} = .03$ $N = 4, K = 2, C = 40$	$M = 320, \text{lr} = .03$ $N = 4, K = 1, C = 40$	$M = 320, \text{lr} = .03$ $N = 4, K = 3, C = 40$
MPC-RHC	$\lambda = 15, \text{lr} = .05$ $N = 8, C = 40$	$\lambda = 60, \text{lr} = .10$ $N = 8, C = 40$	$\lambda = 240, \text{lr} = .05$ $N = 4, C = 20$
MPC- Δt	$\lambda = 30, \text{lr} = .10$	$\lambda = 540, \text{lr} = .10$	$\lambda = 360, \text{lr} = .10$
RHSO	$N = 4, M = 40, \text{lr} = .01$	$N = 4, M = 40, \text{lr} = .01$	$N = 4, M = 40, \text{lr} = .01$

Table 6: **Final JiT hyperparameters: inpainting.** Standard-flow trajectory integrations use Heun with the Euler final-step policy where applicable.

Method	Random inpainting	Box inpainting
PnP-Flow	$N = 20, \gamma_0 = 2 \times 10^5$ $\alpha = .25$	$N = 100, \gamma_0 = 4 \times 10^5$ $\alpha = .25$
D-Flow (1-step)	$M = 320, \text{lr} = .03$ $N = 4, K = 3, C = 40$	$M = 320, \text{lr} = .03$ $N = 4, K = 1, C = 40$
MPC-RHC	$\lambda = 180, \text{lr} = .05$ $N = 4, C = 20$	$\lambda = 30, \text{lr} = .05$ $N = 8, C = 40$
MPC- Δt	$\lambda = 540, \text{lr} = .10$	$\lambda = 180, \text{lr} = .10$
RHSO	$N = 4, M = 40, \text{lr} = .01$	$N = 4, M = 40, \text{lr} = .01$

568 The JiT source-optimization controls and RHSO use the standard-flow solver described above where
569 numerical integration is required. All optimization-based configurations use Adam, and RHSO uses
570 $\mu = 0$.

571 E Full RHSO control-placement results

572 Table 7 gives the complete placement and terminal-depth grid on the standard-flow prior, and table 8
the paired depth-controlled effect of deepening the one-shot planner.

Table 7: **Full JiT control-placement and terminal-depth comparison.** Each cell is PSNR/SSIM/LPIPS. One-shot direct and one-shot suffix use 160 source-state optimizer updates. The two RHSO variants use $N = 4, M = 40$. Best PSNR, best SSIM and lowest LPIPS within each task column are shown in bold, marked independently.

Variant	Denoise	Deblur	Super-res.	Rand.	Box	Eval./BP
One-shot direct	18.19/.376/.713	18.17/.374/.713	18.19/.376/.713	18.18/.375/.713	18.15/.376/.714	161/160
One-shot 4-step suffix	20.91/.502/.553	20.75/.494/.549	20.87/.501/.552	20.92/.502/.548	20.76/.500/.555	1127/1120
RHSO direct	27.79/.734/.198	29.26/.845/.144	27.71/.816/.102	28.20/.835/.094	29.39/.891/.074	167/160
RHSO full suffix	27.43/.698/.267	30.15/.870/.107	27.93/.817/.104	28.58/.847/.075	29.87/.906/.051	647/640

Table 8: **Depth-controlled one-shot JiT comparison.** Differences are 4-step suffix one-shot minus direct one-shot at the same 160 source-state optimizer updates, with paired bootstrap confidence intervals.

Task	Δ PSNR (dB)	Δ LPIPS
Denoise	+2.72 [+2.40,+3.05]	-0.160 [-0.179,-0.140]
Deblur	+2.57 [+2.28,+2.88]	-0.164 [-0.184,-0.145]
Super-res.	+2.69 [+2.36,+3.03]	-0.161 [-0.181,-0.141]
Rand.	+2.75 [+2.44,+3.07]	-0.165 [-0.185,-0.146]
Box	+2.61 [+2.32,+2.93]	-0.159 [-0.179,-0.140]

The suffix control closes the main depth confound in the earlier $N = 1$ versus $N = 4$ study. Numerical depth contributes materially to one-shot JiT quality, but repeated intermediate-state optimization remains much stronger. The full-suffix and direct-planner RHSO rows should not be interpreted as a controlled comparison of endpoint-prediction accuracy alone, because the two planners induce different state updates throughout optimization. Their small average quality difference is nevertheless useful evidence that receding-horizon correction makes RHSO comparatively robust to terminal-planner fidelity.

F Stage-count sweeps

F.1 Fixed total optimizer budget

Tables 9 and 10 sweep the stage count with the total optimizer budget held at $B = NM = 160$, so a larger N redistributes the same number of updates over more stages.

Table 9: **Fixed-total-budget RHSO sweep.** $B = NM = 160$. Each cell is PSNR/SSIM/LPIPS. Best PSNR, best SSIM and lowest LPIPS within each row are shown in bold, marked independently; $\dagger$ marks the frozen default.

Prior	Task	$N = 1$	$N = 2$	$N = 4^\dagger$	$N = 8$	$N = 16$	$N = 32$
JiT	Denoise	18.20/0.376/0.713	27.20/ 0.764 /0.228	27.76 /0.734/ 0.201	26.17/0.652/0.332	25.72/0.639/0.357	25.28/0.620/0.385
JiT	Deblur	18.18/0.375/0.713	26.14/0.730/0.291	29.27 / 0.845 / 0.144	28.55/0.795/0.159	27.95/0.766/0.187	27.49/0.743/0.208
JiT	Super-res.	18.20/0.376/0.713	26.57/0.756/0.222	27.71 / 0.816 / 0.102	26.16/0.746/0.172	25.34/0.709/0.214	24.86/0.674/0.251
JiT	Rand.	18.19/0.375/0.713	26.53/0.752/0.227	28.21 / 0.836 / 0.092	26.76/0.789/0.125	26.33/0.766/0.146	25.75/0.735/0.174
JiT	Box	18.16/0.376/0.714	26.36/0.764/0.242	29.41 / 0.891 / 0.073	28.63/0.861/0.089	28.24/0.843/0.114	27.71/0.810/0.160
pMF	Denoise	23.16/0.592/0.373	29.15 / 0.817 / 0.108	28.50/0.784/0.151	27.81/0.746/0.205	26.98/0.691/0.289	26.22/0.634/0.329
pMF	Deblur	20.94/0.492/0.448	27.44/0.784/0.208	29.90/0.858/0.107	29.94/0.857/0.098	29.95 / 0.859 / 0.093	29.82/0.846/0.099
pMF	Super-res.	21.04/0.503/0.457	27.13/0.791/0.144	27.62 / 0.820 / 0.098	27.19/0.796/0.123	26.83/0.780/0.138	26.60/0.760/0.150
pMF	Rand.	20.89/0.495/0.466	27.27/0.789/0.152	28.24 / 0.838 / 0.082	27.99/0.834/ 0.078	27.79/0.829/0.079	27.52/0.814/0.086
pMF	Box	21.77/0.550/0.410	27.66/0.849/0.122	27.99/0.893/0.069	27.96/ 0.897 / 0.059	27.98/0.893/0.061	28.37 /0.880/0.069

Table 10: **Measurement RMSE for the fixed-total-budget sweep.** Lower is better; these are aggregate diagnostics. Lowest measurement RMSE within each row is shown in bold; $\dagger$ marks the frozen default.

Prior	Task	$N = 1$	$N = 2$	$N = 4^\dagger$	$N = 8$	$N = 16$	$N = 32$
JiT	Denoise	.3352	.2226	.1978	.2005	.2068	.2123
JiT	Deblur	.2378	.0762	.0570	.0678	.0821	.0927
JiT	Super-res.	.2733	.1023	.0586	.0698	.0839	.0996
JiT	Rand.	.1474	.0503	.0242	.0331	.0415	.0495
JiT	Box	.2690	.1100	.0744	.0873	.0995	.1109
pMF	Denoise	.2612	.2036	.1984	.1965	.1946	.1596
pMF	Deblur	.1620	.0624	.0515	.0504	.0531	.0520
pMF	Super-res.	.2053	.0811	.0519	.0487	.0471	.0374
pMF	Rand.	.1106	.0379	.0194	.0164	.0158	.0124
pMF	Box	.1805	.0807	.0696	.0693	.0704	.0538

At fixed $B = 160$, the empirical optimum is finite and model/task dependent. JiT peaks at $N = 4$ in PSNR on all five tasks and then deteriorates through $N = 32$. pMF is more heterogeneous: denoising, super-resolution, and random inpainting lose reconstruction quality at high N while measurement fit can continue improving; deblurring largely saturates; box inpainting shows a high- N PSNR rebound without corresponding SSIM or LPIPS improvement.

F.2 Fixed per-stage optimizer budget

Tables 11 to 13 repeat the sweep with the per-stage budget held at $M = 40$, so the total optimizer work $B = 40N$ grows with the stage count.

With $M = 40$, measurement RMSE improves from $N = 4$ to $N = 16$ for every model-task pair. JiT nevertheless loses PSNR, SSIM, and LPIPS on every task. pMF loses all three metrics on denoising, deblurring, super-resolution, and random inpainting; box inpainting is the global-metric

Table 11: **Fixed-per-stage-budget RHSO sweep.** $M = 40$ and $B = 40N$. Each cell is PSNR/SSIM/LPIPS. Best PSNR, best SSIM and lowest LPIPS within each row are shown in bold, marked independently; † marks the frozen default.

Prior	Task	$N = 1$	$N = 2$	$N = 4^\dagger$	$N = 8$	$N = 16$
JiT	Denoise	16.00/0.323/0.788	23.33/0.623/0.402	27.76/0.734/0.201	26.62/0.654/0.325	26.30/0.634/0.336
JiT	Deblur	15.99/0.323/0.788	22.95/0.599/0.432	29.27/0.845/0.144	27.94/0.761/0.199	25.20/0.625/0.419
JiT	Super-res.	16.00/0.323/0.788	23.21/0.619/0.397	27.71/0.816/0.102	25.69/0.717/0.194	22.76/0.565/0.385
JiT	Rand.	16.00/0.322/0.788	23.17/0.619/0.397	28.21/0.836/0.092	26.58/0.786/0.116	24.84/0.723/0.192
JiT	Box	15.99/0.323/0.788	22.97/0.617/0.410	29.41/0.891/0.073	29.17/0.883/0.071	28.41/0.869/0.094
pMF	Denoise	16.95/0.381/0.588	27.17/0.762/0.185	28.38/0.780/0.153	27.40/0.711/0.255	26.96/0.679/0.310
pMF	Deblur	17.04/0.346/0.583	23.72/0.647/0.346	29.90/0.858/0.107	29.02/0.827/0.131	27.04/0.735/0.305
pMF	Super-res.	17.00/0.354/0.593	23.99/0.671/0.287	27.58/0.819/0.100	26.01/0.749/0.170	23.54/0.639/0.321
pMF	Rand.	16.94/0.352/0.589	24.04/0.670/0.296	28.24/0.838/0.082	27.08/0.814/0.098	25.68/0.773/0.148
pMF	Box	17.17/0.369/0.575	25.14/0.750/0.228	28.02/0.893/0.070	27.92/0.905/0.051	27.03/0.900/0.064

Table 12: **Measurement RMSE for the fixed-per-stage-budget sweep.** Lower is better; these are aggregate diagnostics. Lowest measurement RMSE within each row is shown in bold; † marks the frozen default.

Prior	Task	$N = 1$	$N = 2$	$N = 4^\dagger$	$N = 8$	$N = 16$
JiT	Denoise	.3932	.2555	.1978	.1916	.1917
JiT	Deblur	.3138	.1247	.0570	.0523	.0523
JiT	Super-res.	.3420	.1653	.0586	.0500	.0501
JiT	Rand.	.1856	.0860	.0242	.0194	.0199
JiT	Box	.3363	.1639	.0744	.0690	.0701
pMF	Denoise	.3904	.2192	.1978	.1936	.1919
pMF	Deblur	.2781	.1107	.0515	.0485	.0477
pMF	Super-res.	.3160	.1460	.0523	.0435	.0412
pMF	Rand.	.1704	.0761	.0194	.0138	.0128
pMF	Box	.3143	.1165	.0698	.0671	.0663

Table 13: **Principal paired effects for the fixed- M sweep.** Differences are $N = 16 - N = 4$ with 95% paired bootstrap confidence intervals over 100 images. Positive LPIPS differences are worse.

Prior	Task	Δ PSNR (dB)	Δ LPIPS
JiT	Denoise	-1.46 [-1.73,-1.17]	+0.135 [+0.120,+0.151]
JiT	Deblur	-4.07 [-4.66,-3.49]	+0.275 [+0.236,+0.315]
JiT	Super-res.	-4.95 [-5.33,-4.58]	+0.283 [+0.260,+0.306]
JiT	Rand.	-3.38 [-3.70,-3.07]	+0.100 [+0.083,+0.119]
JiT	Box	-1.00 [-1.42,-0.57]	+0.021 [+0.009,+0.034]
pMF	Denoise	-1.43 [-1.76,-1.12]	+0.157 [+0.130,+0.185]
pMF	Deblur	-2.86 [-3.37,-2.34]	+0.197 [+0.161,+0.234]
pMF	Super-res.	-4.04 [-4.42,-3.64]	+0.220 [+0.199,+0.241]
pMF	Rand.	-2.56 [-2.83,-2.26]	+0.066 [+0.053,+0.079]
pMF	Box	-0.99 [-1.43,-0.53]	-0.006 [-0.013,+0.002]

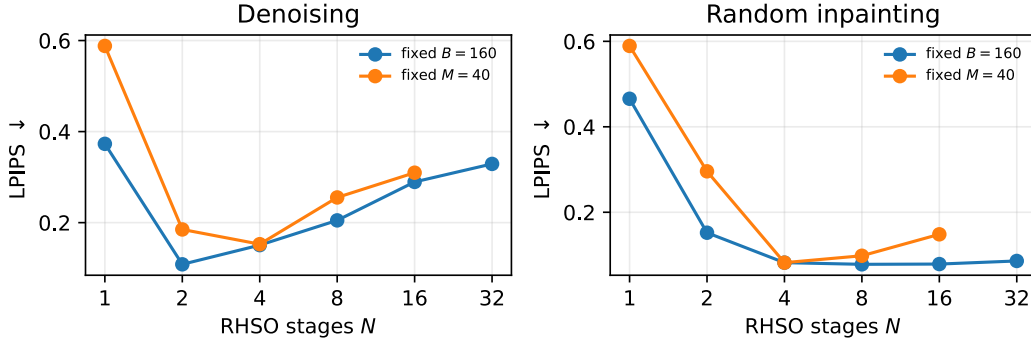


Figure 4: **Fixed-total-budget versus fixed-per-stage-budget pMF sweeps.** Holding $M = 40$ removes optimizer starvation as a primary explanation for the high- N perceptual decline.

exception, though its missing-region reconstruction worsens. This joint fixed- B /fixed- M result is the main reason we reject optimizer starvation as the primary explanation of the high-stage-count reconstruction decline (Section 4.5). The same two sweeps supply the cross-prior comparison of figure 3: reading each stage count as a trajectory budget, the MeanFlow prior wins 86 of 90 task-metric comparisons at fixed B and 71 of 75 at fixed M , including every SSIM and LPIPS comparison, with all eight exceptions in PSNR and six of them on box inpainting.

G Controlled measurement-noise experiment

The experiment uses denoising only and varies $\sigma \in \{0, .1, .2, .4\}$ at fixed $B = 160$. Measurements are exactly paired across N within each noise level and shared across JiT/pMF for a given image and σ . The clean source images and generative seeds are also shared across σ . A protocol nuance is that the measurement seed includes the problem-parameter key, so the standardized Gaussian noise realization is not the same vector rescaled across different σ values. Cross- σ comparisons are therefore distribution-level rather than perfectly pairwise scaled-noise comparisons.

Table 14: **Controlled denoising-noise sweep at $B = 160$.** Each cell is PSNR/SSIM/LPIPS. Best PSNR, best SSIM and lowest LPIPS within each row are shown in bold, marked independently; † marks the frozen default.

Prior	σ	$N = 1$	$N = 2$	$N = 4^\dagger$	$N = 8$	$N = 16$
JiT	0	18.20/0.376/0.713	27.42/0.772/0.238	32.49/0.908/0.069	32.47/0.886/0.077	32.00/0.873/0.090
JiT	0.1	18.20/0.376/0.713	27.37/0.771/0.235	31.36/0.883/0.059	29.87/0.820/0.138	28.93/0.784/0.195
JiT	0.2	18.20/0.376/0.713	27.22/ 0.765/0.226	27.76/0.734/0.202	26.15/0.651/0.334	25.69/0.637/0.360
JiT	0.4	18.19/0.376/0.713	25.88/0.695/0.248	22.00/0.448/0.591	21.45/0.425/0.651	21.28/0.426/0.648
pMF	0	23.00/0.589/0.372	31.34/0.874/0.094	32.57/0.901/0.064	32.71/0.905/0.056	32.92/0.912/0.048
pMF	0.1	22.76/0.584/0.378	30.81/0.863/0.088	31.40/0.879/0.064	31.30/0.876/ 0.061	30.95/0.863/0.088
pMF	0.2	23.01/0.581/0.377	29.22/0.818/0.107	28.54/0.788/0.147	27.81/0.746/0.205	26.97/0.689/0.295
pMF	0.4	22.68/0.575/0.382	23.99/0.600/0.340	22.12/0.476/0.582	21.81/0.450/0.631	21.62/0.435/0.659

Table 15: **Measurement RMSE for the controlled denoising-noise sweep.** Lower is better. Lowest measurement RMSE within each row is shown in bold; † marks the frozen default.

Prior	σ	$N = 1$	$N = 2$	$N = 4^\dagger$	$N = 8$	$N = 16$
JiT	0	.2690	.1019	.0597	.0757	.0916
JiT	0.1	.2870	.1420	.1124	.1204	.1306
JiT	0.2	.3352	.2224	.1978	.2002	.2070
JiT	0.4	.4822	.4075	.3761	.3773	.3811
pMF	0	.1732	.0625	.0547	.0550	.0547
pMF	0.1	.2061	.1158	.1105	.1102	.1093
pMF	0.2	.2635	.2035	.1982	.1966	.1946
pMF	0.4	.4366	.3863	.3751	.3737	.3724

Table 16: **Noise-dependent value of additional stages.** Differences are $N = 4 - N = 2$ with 95% paired bootstrap confidence intervals.

Prior	σ	Δ PSNR (dB)	Δ LPIPS
JiT	0	+5.07 [+4.71,+5.42]	-0.169 [-0.186,-0.152]
JiT	0.1	+3.99 [+3.67,+4.32]	-0.176 [-0.193,-0.158]
JiT	0.2	+0.54 [+0.07,+1.01]	-0.024 [-0.054,+0.005]
JiT	0.4	-3.89 [-4.40,-3.38]	+0.342 [+0.303,+0.381]
pMF	0	+1.24 [+0.94,+1.55]	-0.030 [-0.038,-0.022]
pMF	0.1	+0.59 [+0.38,+0.80]	-0.024 [-0.032,-0.017]
pMF	0.2	-0.68 [-0.89,-0.48]	+0.040 [+0.027,+0.053]
pMF	0.4	-1.87 [-2.22,-1.54]	+0.242 [+0.208,+0.277]

For pMF, the reconstruction-optimal stage count shifts from $N = 16$ at $\sigma = 0$ to approximately $N = 4$ to 8 at .1, then to $N = 2$ at .2 and .4. JiT remains strongest near $N = 4$ at low and moderate noise but also moves to $N = 2$ at $\sigma = .4$. The sign reversal in table 16 is robust at the image level and is the strongest direct evidence that aggressive distributed control can fit noise rather than the clean target.

H State-anchor experiment

Tables 17 to 19 sweep the anchor weight μ of equation (12) for denoising at $\sigma = .2$, holding the rest of the frozen configuration fixed.

Table 17: **State-anchor sweep for denoising at $\sigma = .2$.** Each cell is PSNR/SSIM/LPIPS. Best PSNR, best SSIM and lowest LPIPS within each row are shown in bold, marked independently; $\dagger$ marks the frozen default.

Prior	N	$\mu = 0^\dagger$	$\mu = 0.01$	$\mu = 0.03$	$\mu = 0.1$	$\mu = 0.3$	$\mu = 1$
JiT	4	27.76/0.734/0.201	27.83/0.735/0.199	27.88/0.738/0.194	28.27/0.757/0.172	29.15/0.808/0.105	28.65/0.802/0.145
JiT	8	26.17/0.652/0.332	26.13/0.650/0.333	26.21/0.653/0.328	26.35/0.662/0.315	26.79/0.688/0.275	28.29/0.776/0.147
pMF	4	28.41/0.781/0.155	28.44/0.778/0.162	28.32/0.774/0.168	28.11/0.758/0.198	28.52/0.773/0.174	29.47/0.826/0.083
pMF	8	27.74/0.739/0.212	27.61/0.728/0.222	27.63/0.734/0.223	27.59/0.724/0.238	27.63/0.721/0.249	28.71/0.780/0.164

Table 18: **Measurement RMSE for the state-anchor sweep.** Lower is better. Lowest measurement RMSE within each row is shown in bold; $\dagger$ marks the frozen default.

Prior	N	$\mu = 0^\dagger$	$\mu = 0.01$	$\mu = 0.03$	$\mu = 0.1$	$\mu = 0.3$	$\mu = 1$
JiT	4	.1978	.1976	.1978	.1985	.2025	.2128
JiT	8	.2005	.2008	.2006	.2004	.2014	.2080
pMF	4	.1981	.1976	.1974	.1964	.1973	.2065
pMF	8	.1965	.1960	.1956	.1948	.1945	.1989

Table 19: **Paired effect of the strongest anchor setting in the μ ablation.** Differences are anchored minus $\mu = 0$ with 95% paired bootstrap confidence intervals.

Prior	N	anchor	Δ PSNR (dB)	Δ LPIPS
JiT	4	$\mu = 0.3$	+1.38 [+1.19,+1.58]	-0.096 [-0.110,-0.082]
JiT	8	$\mu = 1$	+2.12 [+1.80,+2.44]	-0.185 [-0.209,-0.162]
pMF	4	$\mu = 1$	+1.05 [+0.81,+1.31]	-0.072 [-0.087,-0.058]
pMF	8	$\mu = 1$	+0.96 [+0.73,+1.21]	-0.048 [-0.066,-0.030]

Strong anchoring improves reconstruction while relaxing measurement fit, but the response is not monotone in μ . In particular, pMF at low and intermediate anchor values can reduce measurement RMSE while worsening LPIPS before the stronger $\mu = 1$ condition reverses the perceptual trend. The pMF $N = 4$, $\mu = 1$ denoising condition reaches 29.47/.826/.083, better than the frozen $\mu = 0$ headline denoising result. We do not substitute this result into the main benchmark because the anchor grid was evaluated on the same 100-image test set; doing so would be post-hoc test-set hyperparameter selection.

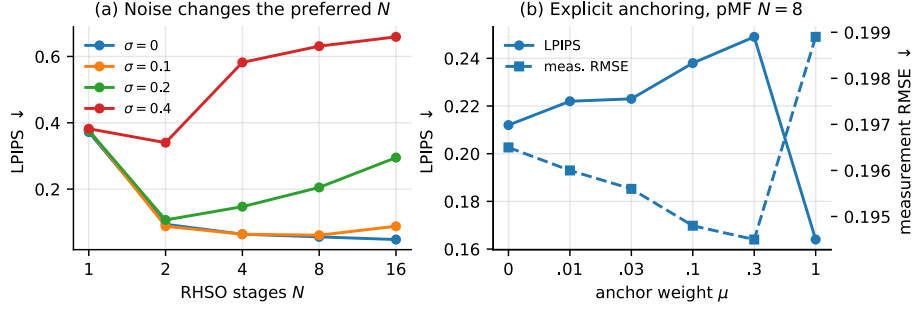


Figure 5: **Distributed control exposes a data-fit versus reconstruction tradeoff.** (a) In pMF denoising, many stages help when the measurement is clean and hurt increasingly as noise grows. (b) At pMF $N = 8$, strong anchoring reverses the LPIPS degradation while giving back measurement fit. Values are those of tables 14, 17 and 18; the two panels summarize Appendices G and H.

I Temporal allocation and initialization

This appendix contains the temporal-placement study referenced in Section 4.5. It is reported separately from the main control experiments because the schedule result is task dependent and does not bear on the data-fit versus reconstruction tradeoff studied there.

I.1 Schedule exponent

Tables 20 and 21 sweep the schedule exponent β of equation (8) at the frozen $N = 4, M = 40$ budget, where $\beta = 1$ is the uniform grid used in every headline run.

Table 20: **Nonuniform RHSO schedule sweep.** $N = 4, M = 40$. Each cell is PSNR/SSIM/LPIPS. Best PSNR, best SSIM and lowest LPIPS within each row are shown in bold, marked independently; $\dagger$ marks the frozen default.

Prior	Task	$\beta = 0.5$	$\beta = 0.75$	$\beta = 1^\dagger$	$\beta = 1.5$	$\beta = 2$
JiT	Denoise	26.85/0.665/0.329	27.10/0.687/0.291	27.79/0.734/0.198	28.36/0.795/0.143	27.35/0.772/0.201
JiT	Deblur	29.37/0.829/0.128	29.65/0.856/0.115	29.26/0.845/0.144	27.74/0.790/0.222	26.49/0.741/0.275
JiT	Super-res.	27.17/0.784/0.139	27.57/0.807/0.114	27.71/0.816/0.102	27.32/0.797/0.146	26.38/0.762/0.200
JiT	Rand.	28.09/0.837/ 0.071	28.32/0.843/0.076	28.20/0.835/0.094	27.40/0.799/0.153	26.49/0.759/0.210
JiT	Box	29.68/ 0.907/0.044	29.89/0.905/0.047	29.39/0.891/0.074	28.15/0.837/0.154	26.72/0.780/0.221
pMF	Denoise	26.94/0.682/0.307	27.74/0.737/0.219	28.56/0.786/0.150	28.98/0.810/0.118	28.93/0.809/0.122
pMF	Deblur	29.53/0.845/0.107	29.87/0.856/0.098	29.82/0.854/0.113	29.23/0.841/0.147	28.52/0.818/0.174
pMF	Super-res.	26.55/0.778/0.142	27.37/0.808/0.109	27.73/0.820/ 0.096	27.93/0.823/0.102	27.72/0.810/0.123
pMF	Rand.	27.51/0.828/0.086	28.15/0.839/ 0.075	28.30/0.841/0.078	28.30/0.834/0.096	27.92/0.814/0.125
pMF	Box	27.91/ 0.904/0.054	27.91/0.900/0.059	27.96/0.892/0.072	27.59/0.863/0.106	27.37/0.849/0.122

Table 21: **Measurement RMSE for the nonuniform schedule sweep.** Lower is better. Lowest measurement RMSE within each row is shown in bold; $\dagger$ marks the frozen default.

Prior	Task	$\beta = 0.5$	$\beta = 0.75$	$\beta = 1^\dagger$	$\beta = 1.5$	$\beta = 2$
JiT	Denoise	.1904	.1929	.1981	.2094	.2206
JiT	Deblur	.0499	.0526	.0563	.0634	.0783
JiT	Super-res.	.0472	.0515	.0587	.0753	.0989
JiT	Rand.	.0173	.0194	.0245	.0357	.0486
JiT	Box	.0662	.0690	.0751	.0887	.1087
pMF	Denoise	.1920	.1948	.1983	.2024	.2055
pMF	Deblur	.0494	.0503	.0516	.0536	.0559
pMF	Super-res.	.0464	.0486	.0520	.0600	.0714
pMF	Rand.	.0152	.0168	.0191	.0242	.0311
pMF	Box	.0670	.0682	.0698	.0757	.0792

Measurement RMSE increases strictly with β in all ten model-task pairs. The best LPIPS schedule is $\beta = 1.5$ for denoising in both models, .75 for deblurring, 1 for super-resolution, .5 for box inpainting,

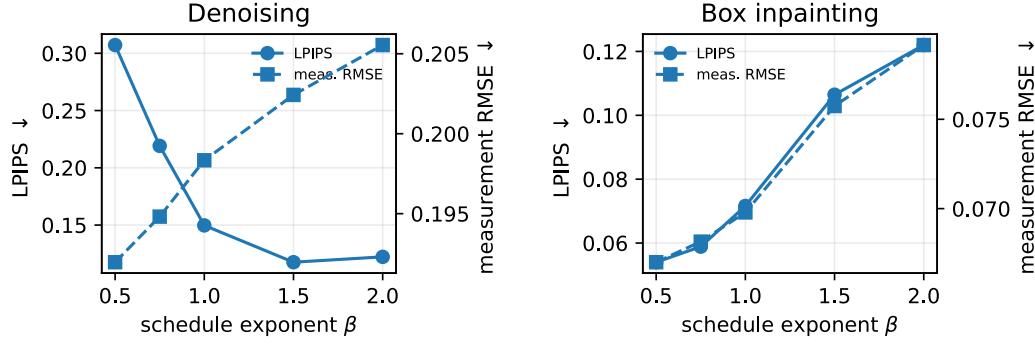


Figure 6: **Task-dependent temporal allocation for pMF.** Denoising prefers earlier/noisy-side control with $\beta > 1$, even though measurement fit worsens, while box inpainting prefers later/data-side control with $\beta < 1$.

633 and .5 JiT / .75 pMF for random inpainting. This agreement across the two priors supports a task-
 634 dependent temporal-allocation interpretation, while the lack of one universal direction motivates
 635 keeping $\beta = 1$ as the task-agnostic frozen benchmark.

636 I.2 Initialization time

637 The canonical time $\tau_{\text{init}} = 0$ denotes pure prior noise. Increasing τ_{init} injects more measurement-
 638 guide content into the starting state.

Table 22: **Canonical initialization-time ablation for RHSO.** Each cell is PSNR/SSIM/LPIPS; $\tau_{\text{init}} = 0$ is pure prior noise and larger values inject more measurement-guide content. Best PSNR, best SSIM and lowest LPIPS within each row are shown in bold, marked independently; $\dagger$ marks the frozen default.

Prior	Task	$\tau = 0^\dagger$	.2	.4	.6
JiT	Denoise	27.79/.734/.198	26.82/.654/.337	26.62/.643/.352	26.44/.632/.356
	Deblur	29.26/.845/.144	30.57/.873/.070	29.62/.827/.120	27.44/.712/.309
	Super-res.	27.71/.816/.102	27.46/.795/.121	26.70/.759/.159	25.47/.694/.238
	Box	29.39/.891/.074	27.95/ .909/.045	26.62/.902/.062	26.41/.892/.088
	Rand.	28.20/.835/.094	28.31/.842/.068	26.08/.792/.139	18.03/.497/.485
pMF	Denoise	28.48/.783/.152	28.15/.762/.186	27.50/.719/.244	26.87/.674/.321
	Deblur	29.95/.861/.103	30.50/.873/.061	29.89/.857/.077	28.72/.812/.159
	Super-res.	27.78/.820/.098	27.52/.804/.109	26.72/.775/.143	25.70/.732/.195
	Box	28.10/.892/.072	27.38/.900/ .063	26.23/ .901/.063	26.33/.899/ .063
	Rand.	28.29/.840/.079	28.15/.838/ .074	27.40/.825/.091	23.90/.752/.205

Table 23: **Measurement RMSE for the initialization-time ablation.** These are aggregate diagnostics. Lowest measurement RMSE within each row is shown in bold; $\dagger$ marks the frozen default.

Prior	Task	$\tau = 0^\dagger$	.2	.4	.6
JiT	Denoise	.1981	.1894	.1884	.1878
JiT	Deblur	.0563	.0499	.0488	.0477
JiT	Super-res.	.0587	.0468	.0439	.0419
JiT	Box	.0751	.0656	.0643	.0637
JiT	Rand.	.0245	.0160	.0144	.0146
pMF	Denoise	.1977	.1961	.1937	.1913
pMF	Deblur	.0508	.0495	.0488	.0482
pMF	Super-res.	.0523	.0471	.0446	.0426
pMF	Box	.0700	.0680	.0669	.0663
pMF	Rand.	.0194	.0159	.0148	.0141

639 Measurement fit generally improves as the start moves toward the data end, but reconstruction often
 640 degrades; deblurring is a clear exception that benefits from a moderate measurement-informed start.

We retain pure-noise initialization in the headline benchmark because it is a simple task-agnostic operating point and demonstrates that the reconstruction quality is produced by the control procedure rather than by favorable image content already present in the starting state.

J Additional quantitative diagnostics

J.1 Measurement consistency for the main solver benchmark

Table 24 reports measurement-space RMSE for the headline configurations of table 1. Measurement fit and reconstruction quality do not order the solvers identically: an MPC variant attains the lowest residual in nine of the ten prior-task pairs, and RHSO is lowest in none, which is why the two families of metric are reported separately throughout.

Table 24: **Measurement-space RMSE for selected headline configurations.** Lower is better; the lowest RMSE per task within each prior block is shown in bold.

Prior	Method	Denoise	Deblur	Super-res.	Rand.	Box
pMF	PnP-Flow	.1745	.0501	.0794	.0383	.0286
	D-Flow (1-step)	.2365	.0906	.1291	.0641	.1302
	MPC-RHC	.2074	.0453	.0918	.0472	.0225
	MPC- Δt	.1759	.0456	.0163	.0053	.0312
	RHSO	.1980	.0510	.0523	.0196	.0697
JiT	PnP-Flow	.1857	.0495	.0721	.0255	.0274
	D-Flow (1-step)	.3153	.2068	.2482	.1323	.2446
	MPC-RHC	.2051	.0454	.0753	.0370	.0228
	MPC- Δt	.1775	.0458	.0155	.0048	.0306
	RHSO suffix	.1929	.0516	.0509	.0185	.0686

J.2 Regional inpainting quality

Table 25 splits inpainting PSNR into the missing and observed regions.

Table 25: **Regional inpainting PSNR for the headline benchmark.** Missing-region PSNR measures reconstruction of unobserved content; observed-region PSNR measures preservation of measured pixels. Highest PSNR per column within each prior block is shown in bold.

Prior	Method	Random inpainting		Box inpainting	
		Missing	Observed	Missing	Observed
pMF	PnP-Flow	23.28	29.20	12.32	33.71
	D-Flow (1-step)	22.89	24.69	12.21	24.29
	MPC-RHC	22.86	27.36	9.50	33.27
	MPC- Δt	24.54	43.29	13.40	34.39
	RHSO	24.57	35.18	12.61	31.48
JiT	PnP-Flow	24.57	32.79	16.07	33.81
	D-Flow (1-step)	18.18	18.37	13.42	18.32
	MPC-RHC	24.58	29.49	11.89	33.28
	MPC- Δt	25.06	43.83	14.51	34.52
	RHSO suffix	24.81	35.69	14.83	31.70

Global inpainting metrics can obscure where an error occurs. This is important in the fixed- M study: pMF box inpainting retains favorable whole-image SSIM/LPIPS at high N even while reconstruction of the missing region worsens. Random inpainting provides a cleaner spatial tradeoff, where high N can improve observed-region fidelity while degrading the missing region.

K Runtime and model-side work

The substantial task dependence for MPC methods follows directly from the selected horizon and control budgets in tables 3 to 6. RHSO uses the same $N = 4$, $M = 40$ structural budget on every task, producing comparatively stable runtime within each model family.

Table 26: **Runtime per image for each selected configuration.** Times are seconds/image at batch size one. The standard-flow RHSO entry is the integrated-suffix planner, as in table 1.

Prior	Method	Denoising	Deblurring	Super-res.	Rand. inpaint.	Box inpaint.
pMF	PnP-Flow	0.81	4.00	0.35	0.34	1.69
	D-Flow (1-step)	26.97	35.01	25.49	25.37	27.35
	MPC-RHC	14.33	4.51	26.52	26.39	1.17
	MPC- Δt	21.71	29.83	4.60	4.51	22.20
	RHSO	12.97	17.98	13.23	13.19	14.01
JiT	PnP-Flow	1.85	3.83	0.76	0.77	3.66
	D-Flow (1-step)	27.98	28.31	27.85	27.93	28.21
	MPC-RHC	14.13	0.49	28.01	28.03	0.24
	MPC- Δt	24.72	25.17	5.38	5.39	24.73
	RHSO suffix	56.68	56.91	56.56	56.75	56.37

660 K.1 Task-specific model evaluations and backpropagations

Table 27: **Generative-model evaluations/backpropagations per image.** The standard-flow RHSO entry is the integrated-suffix planner; the direct planner costs 167/160 (table 7).

Prior	Method	Denoising	Deblurring	Super-res.	Rand. inpaint.	Box inpaint.
pMF	PnP-Flow	51/0	101/0	21/0	21/0	101/0
	D-Flow (1-step)	321/320	321/320	321/320	321/320	321/320
	MPC-RHC	168/160	8/0	328/320	328/320	8/0
	MPC- Δt	288/280	288/280	64/60	64/60	288/280
	RHSO	164/160	164/160	164/160	164/160	164/160
JiT	PnP-Flow	51/0	101/0	21/0	21/0	101/0
	D-Flow (1-step)	321/320	321/320	321/320	321/320	321/320
	MPC-RHC	164/160	4/0	324/320	324/320	4/0
	MPC- Δt	288/280	288/280	64/60	64/60	288/280
	RHSO suffix	647/640	647/640	647/640	647/640	647/640

661 The MPC-RHC $K = 1$ configurations require no generative-model backpropagation through the
662 optimized control objective because the nominal first transition can be cached. In contrast, $K = 3$
663 requires repeated differentiation through two subsequent nominal transitions.

664 K.2 Sampling-step reference and runtime caveats

665 For reference, a plain SDEdit-style partial-noising pass [50] performs no state optimization at all
666 and therefore costs a handful of model evaluations with no backpropagation; every solver in table 1
667 that differentiates the prior is orders of magnitude more expensive in backpropagations, which is
668 the cost RHSO trades for reconstruction quality. The task-specific wall-clock measurements above
669 use batch size one and the frozen headline implementations. They should not be interpreted as
670 architecture-controlled hardware comparisons because pMF and JiT differ in model size, framework,
671 and implementation. Eval./BP counts are the cleaner statement about algorithmic terminal-planning
672 depth.

673 L Additional qualitative results, limitations, and broader impact

674 Figure 2 shows one set of examples from the frozen pMF benchmark. Figure 7 shows a second,
675 independently drawn set with the same column layout. Both are illustrative and do not by themselves
676 establish perceptual superiority; the aggregate evidence is table 1.

677 **Limitations.** First, pMF-L/16 and JiT-B/16 differ in model capacity, training and guidance scale,
678 so cross-prior quality and runtime differences are system-level and cannot isolate the MeanFlow
679 training objective. Second, the Gaussian theory assumes a linear covariance-aligned transport and an
680 idealized additive control allocation; actual RHSO is nonlinear and sequential. Third, the controlled-
681 noise sweep uses independently seeded Gaussian perturbations across noise levels rather than one

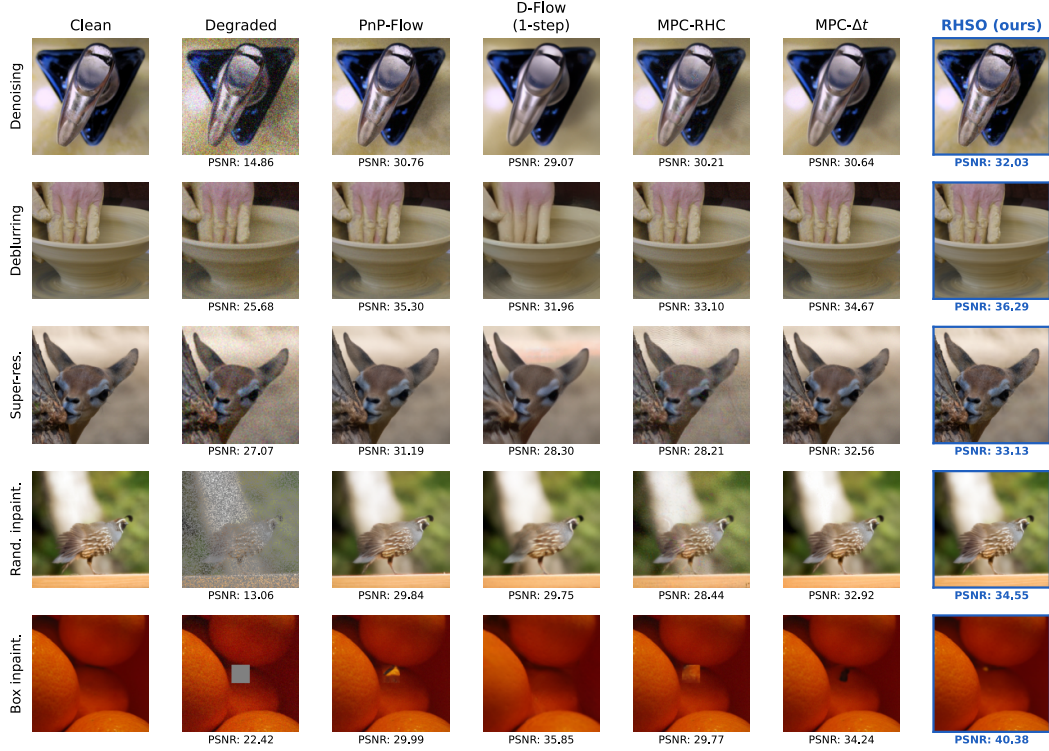


Figure 7: A second qualitative comparison for the five inverse problems. Columns show the clean target, degraded observation, and pMF solver outputs from the frozen benchmark, on images distinct from those of figure 2.

standardized perturbation scaled pathwise. Fourth, the strongest anchor and nonuniform schedule values were identified on the same 100-image evaluation set and are therefore mechanistic ablations, not replacements for the frozen headline hyperparameters. Fifth, all evaluations assume access to the ground-truth ImageNet source class for conditioning; class-agnostic or estimated-class reconstruction remains untested. The evaluation is limited to 100 images and image-level metrics rather than distributional perceptual measures. Finally, the few-step comparison of Section 4.3 sweeps N for a single solver family (RHSO) under two budget controls; it shows the pMF system is more robust than the JiT system as this trajectory budget shrinks, but not that every solver inherits the same advantage or that the effect is unique to MeanFlow among all few-step-capable priors. These restrictions are reflected in the claim language throughout the paper.

Broader impact. Generative priors can improve recovery from incomplete or noisy measurements and may be useful in scientific and computational imaging. The same prior can also synthesize plausible content that is not uniquely supported by the observation, particularly in severely underdetermined settings. Reconstructions from methods such as RHSO should therefore not be treated as verified ground truth in high-stakes imaging without task-specific validation, uncertainty analysis, and safeguards appropriate to the deployment context.

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Justification: Appendix C.4 states the procedure: paired 95% nonparametric bootstrap confidence intervals over the 100 per-image differences at a fixed seed, with pairing preserved across image, class, operator, measurement realization and initial noise. Intervals are

reported for every principal paired effect, in Tables 8, 13, 16 and 19. The benchmark and sweep tables report point estimates over that same fixed 100-image set; quantities available only as job-level aggregates—measurement RMSE in the compact archives—are identified as such in Appendix C.4 and are not treated as 100 independent samples.

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